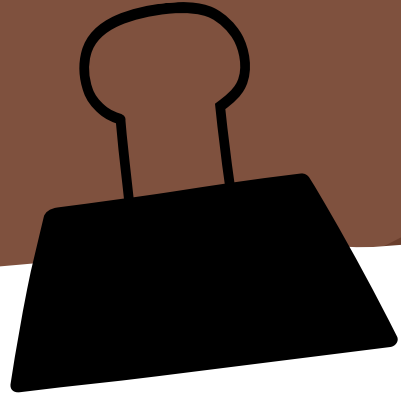




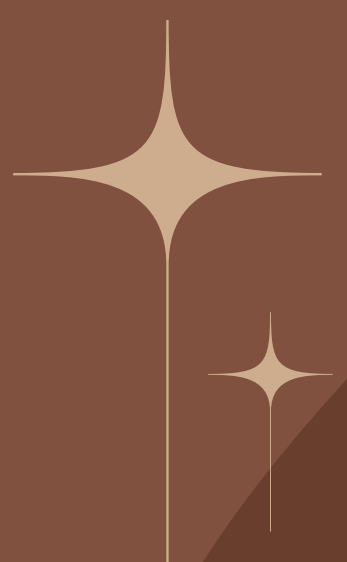
END TO END DATA ANALYTICS PROJECT

(Python + MySQL + Power BI)

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ABOUT

END-TO-END DATA ANALYTICS PROJECT
ANALYZING 13K+ RETAIL TRANSACTIONS
USING PYTHON, MYSQL, AND POWER BI
TO UNCOVER SALES TRENDS, CUSTOMER
BEHAVIOR, AND REVENUE INSIGHTS.



PROBLEM STATEMENT



PROJECT OBJECTIVE:

TO BUILD A ROBUST DATA PIPELINE THAT CLEANSSES
RAW, INCONSISTENT TRANSACTIONAL DATA, STORES IT
IN A RELATIONAL DATABASE, AND DELIVERS AN
INTERACTIVE POWER BI DASHBOARD TO OPTIMIZE
INVENTORY AND MARKETING STRATEGIES.



DATA CLEANING IN PYTHON AND STORING THE DATA IN SQL DATABASE

Importing python libraiaies for data cleaning

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading the dataset

```
In [3]: df = pd.read_csv("C:/Users/mmutt/OneDrive/Desktop/practice.py/Python + Power BI ETE Project/retail_store_sales.csv")
```

Exploring the data

```
In [4]: df.head()
```

```
Out[4]:
```

	Transaction ID	Customer ID	Category	Item	Price Per Unit	Quantity	Total Spent	Payment Method	Location	Transaction Date
0	TXN_7331955	CUST_12	Milk Products	Item_4_MILK	9.5	7.0	66.5	Credit Card	Online	01-01-2022
1	TXN_6950467	CUST_01	Computers and electric accessories	Item_6_CEA	12.5	6.0	75.0	Digital Wallet	In-store	01-01-2022
2	TXN_1519499	CUST_15	Computers and electric accessories	Item_6_CEA	12.5	6.0	75.0	Credit Card	Online	01-01-2022
3	TXN_2070726	CUST_22	Milk Products	Item_6_MILK	12.5	5.0	62.5	Digital Wallet	In-store	01-01-2022
4	TXN_9364474	CUST_19	Food	Item_22_FOOD	36.5	3.0	109.5	Cash	In-store	01-01-2022

```
In [5]: df.duplicated().sum()
```

```
Out[5]: np.int64(0)
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: Transaction ID      0
Customer ID      0
Category      0
Item      1213
Price Per Unit      609
Quantity      604
Total Spent      604
Payment Method      0
Location      0
Transaction Date      0
dtype: int64
```

Item column have some null values, lets fill these by mode value after grouping based on category

```
In [7]: df.columns = df.columns.str.lower().str.replace(' ', '_')
df.columns
```

```
Out[7]: Index(['transaction_id', 'customer_id', 'category', 'item', 'price_per_unit',
              'quantity', 'total_spent', 'payment_method', 'location',
              'transaction_date'],
              dtype='object')
```

converted the columns name to lower case

```
In [8]: df['item'] = df.groupby('category')['item'].transform(lambda x: x.fillna(x.mode()[0]))
```

```
In [9]: df['item'].isnull().sum()
```

```
Out[9]: np.int64(0)
```

we have some null values in 'price_per_unit', 'quantity', 'total_spent', first lets check total_spent column is correct or not

```
In [10]: (df['price_per_unit'] * df['quantity'] == df['total_spent']).all()
```

```
Out[10]: np.False_
```

it means total spent column is not correct so lets drop it and make a new column

```
In [11]: df.drop(columns = 'total_spent', inplace = True)
```

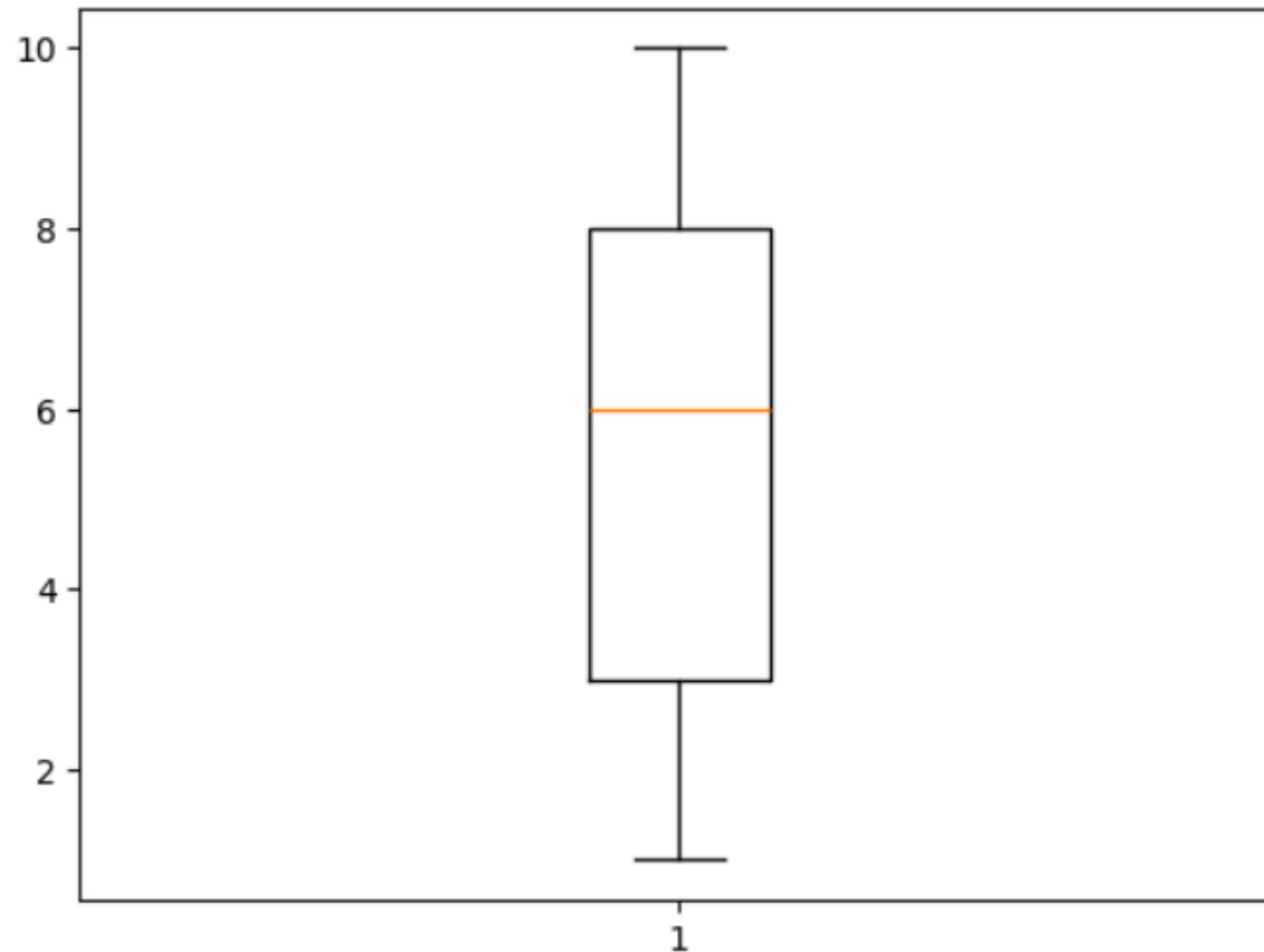
lets fill the null values in price_per_unit and quantity

```
In [12]: df['price_per_unit'] = df.groupby('item')['price_per_unit'].transform(lambda x: x.fillna(x.median()))
```

```
In [13]: df['price_per_unit'].isnull().sum()
```

```
Out[13]: np.int64(0)
```

```
In [14]: plt.boxplot(df['quantity'].dropna())  
plt.show()
```



since the data does not contains outliers so filling with mode would be better

```
In [15]: df['quantity'] = df.groupby('item')['quantity'].transform(lambda x: x.fillna(x.mode()[0]))
```

```
In [16]: df.isnull().sum()
```

```
Out[16]: transaction_id    0
customer_id    0
category    0
item    0
price_per_unit    0
quantity    0
payment_method    0
location    0
transaction_date    0
dtype: int64
```

```
In [17]: df['total_spent'] = df['price_per_unit']*df['quantity']
df.head()
```

```
Out[17]:
```

	transaction_id	customer_id	category	item	price_per_unit	quantity	payment_method	location	transaction_date	total_spent
0	TXN_7331955	CUST_12	Milk Products	Item_4_MILK	9.5	7.0	Credit Card	Online	01-01-2022	66.5
1	TXN_6950467	CUST_01	Computers and electric accessories	Item_6_CEA	12.5	6.0	Digital Wallet	In-store	01-01-2022	75.0
2	TXN_1519499	CUST_15	Computers and electric accessories	Item_6_CEA	12.5	6.0	Credit Card	Online	01-01-2022	75.0
3	TXN_2070726	CUST_22	Milk Products	Item_6_MILK	12.5	5.0	Digital Wallet	In-store	01-01-2022	62.5
4	TXN_9364474	CUST_19	Food	Item_22_FOOD	36.5	3.0	Cash	In-store	01-01-2022	109.5

```
In [18]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 12575 entries, 0 to 12574  
Data columns (total 10 columns):  
#   Column                Non-Null Count  Dtype    
---  ---                  
0   transaction_id         12575 non-null  object   
1   customer_id           12575 non-null  object   
2   category              12575 non-null  object   
3   item                  12575 non-null  object   
4   price_per_unit        12575 non-null  float64  
5   quantity              12575 non-null  float64  
6   payment_method        12575 non-null  object   
7   location               12575 non-null  object   
8   transaction_date       12575 non-null  object   
9   total_spent            12575 non-null  float64  
dtypes: float64(3), object(7)  
memory usage: 982.6+ KB
```

transaction_date datatype is incorrect

```
In [19]: df['transaction_date'] = pd.to_datetime(df['transaction_date'], format = '%d-%m-%Y')  
df['transaction_date'].dtype
```

```
Out[19]: dtype('<M8[ns]>')
```

Creating more columns for analysis

```
In [20]: df['month'] = df['transaction_date'].dt.month_name()  
df['day'] = df['transaction_date'].dt.day_name()  
  
df.head()
```

```
Out[20]:
```

	transaction_id	customer_id	category	item	price_per_unit	quantity	payment_method	location	transaction_date	total_spent	month	day
0	TXN_7331955	CUST_12	Milk Products	Item_4_MILK	9.5	7.0	Credit Card	Online	2022-01-01	66.5	January	Saturday
1	TXN_6950467	CUST_01	Computers and electric accessories	Item_6_CEA	12.5	6.0	Digital Wallet	In-store	2022-01-01	75.0	January	Saturday
2	TXN_1519499	CUST_15	Computers and electric accessories	Item_6_CEA	12.5	6.0	Credit Card	Online	2022-01-01	75.0	January	Saturday
3	TXN_2070726	CUST_22	Milk Products	Item_6_MILK	12.5	5.0	Digital Wallet	In-store	2022-01-01	62.5	January	Saturday
4	TXN_9364474	CUST_19	Food	Item_22_FOOD	36.5	3.0	Cash	In-store	2022-01-01	109.5	January	Saturday

```
In [21]: def day_type(value):
        if value in ['Saturday', 'Sunday']:
            return 'Weekend'
        else:
            return 'Weekday'
df['day_type'] = df['day'].apply(day_type)
df.head()
```

Out[21]:

	transaction_id	customer_id	category	item	price_per_unit	quantity	payment_method	location	transaction_date	total_spent	month	day	day_type
0	TXN_7331955	CUST_12	Milk Products	Item_4_MILK	9.5	7.0	Credit Card	Online	2022-01-01	66.5	January	Saturday	Weekend
1	TXN_6950467	CUST_01	Computers and electric accessories	Item_6_CEA	12.5	6.0	Digital Wallet	In-store	2022-01-01	75.0	January	Saturday	Weekend
2	TXN_1519499	CUST_15	Computers and electric accessories	Item_6_CEA	12.5	6.0	Credit Card	Online	2022-01-01	75.0	January	Saturday	Weekend
3	TXN_2070726	CUST_22	Milk Products	Item_6_MILK	12.5	5.0	Digital Wallet	In-store	2022-01-01	62.5	January	Saturday	Weekend
4	TXN_9364474	CUST_19	Food	Item_22_FOOD	36.5	3.0	Cash	In-store	2022-01-01	109.5	January	Saturday	Weekend

Now the dataset is cleaned and ready to use for building dashboard and answering business questions

```
In [26]: from sqlalchemy import create_engine
```

```
In [27]: engine = create_engine('mysql+pymysql://root:6395@localhost/salesdb')
```

created engine for connection to mysql

```
In [28]: df.to_sql(name = 'sales_data', con = engine, index = False, if_exists = 'replace')
```

```
Out[28]: 12575
```

now the dataset is stored in mysql database

In [29]: `pd.read_sql_query('SELECT * FROM sales_data', engine)`

Out[29]:		transaction_id	customer_id	category	item	price_per_unit	quantity	payment_method	location	transaction_date	total_spent	month	day	day_type
	0	TXN_7331955	CUST_12	Milk Products	Item_4_MILK	9.5	7.0	Credit Card	Online	2022-01-01	66.5	January	Saturday	Weekend
	1	TXN_6950467	CUST_01	Computers and electric accessories	Item_6_CEA	12.5	6.0	Digital Wallet	In-store	2022-01-01	75.0	January	Saturday	Weekend
	2	TXN_1519499	CUST_15	Computers and electric accessories	Item_6_CEA	12.5	6.0	Credit Card	Online	2022-01-01	75.0	January	Saturday	Weekend
	3	TXN_2070726	CUST_22	Milk Products	Item_6_MILK	12.5	5.0	Digital Wallet	In-store	2022-01-01	62.5	January	Saturday	Weekend
	4	TXN_9364474	CUST_19	Food	Item_22_FOOD	36.5	3.0	Cash	In-store	2022-01-01	109.5	January	Saturday	Weekend
	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	12570	TXN_1505827	CUST_13	Computers and electric accessories	Item_19_CEA	32.0	3.0	Digital Wallet	Online	2025-01-18	96.0	January	Saturday	Weekend
	12571	TXN_5907338	CUST_25	Beverages	Item_23_BEV	38.0	9.0	Cash	In-store	2025-01-18	342.0	January	Saturday	Weekend
	12572	TXN_5804265	CUST_11	Computers and electric accessories	Item_20_CEA	33.5	2.0	Digital Wallet	Online	2025-01-18	67.0	January	Saturday	Weekend
	12573	TXN_1661883	CUST_04	Beverages	Item_17_BEV	29.0	9.0	Cash	In-store	2025-01-18	261.0	January	Saturday	Weekend
	12574	TXN_5709336	CUST_10	Food	Item_4_FOOD	9.5	2.0	Credit Card	In-store	2025-01-18	19.0	January	Saturday	Weekend

12575 rows × 13 columns

Now the dataset is ready for building dashboard



POWER BI DASHBOARD TO UNCOVER INSIGHTS

Month

All

In-store

Online

Cash

Credit Card

Digital Wallet

13K

Total transactions

69.15K

Total quantity sold

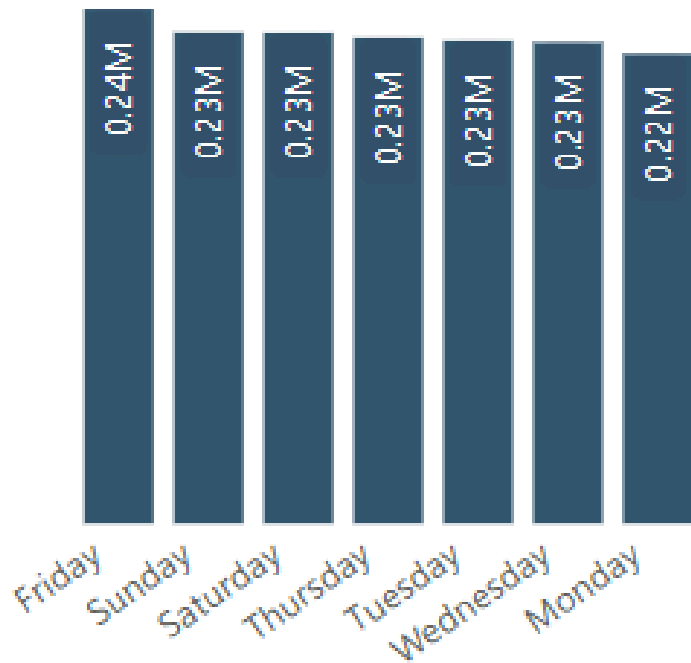
1.63M

Total revenue

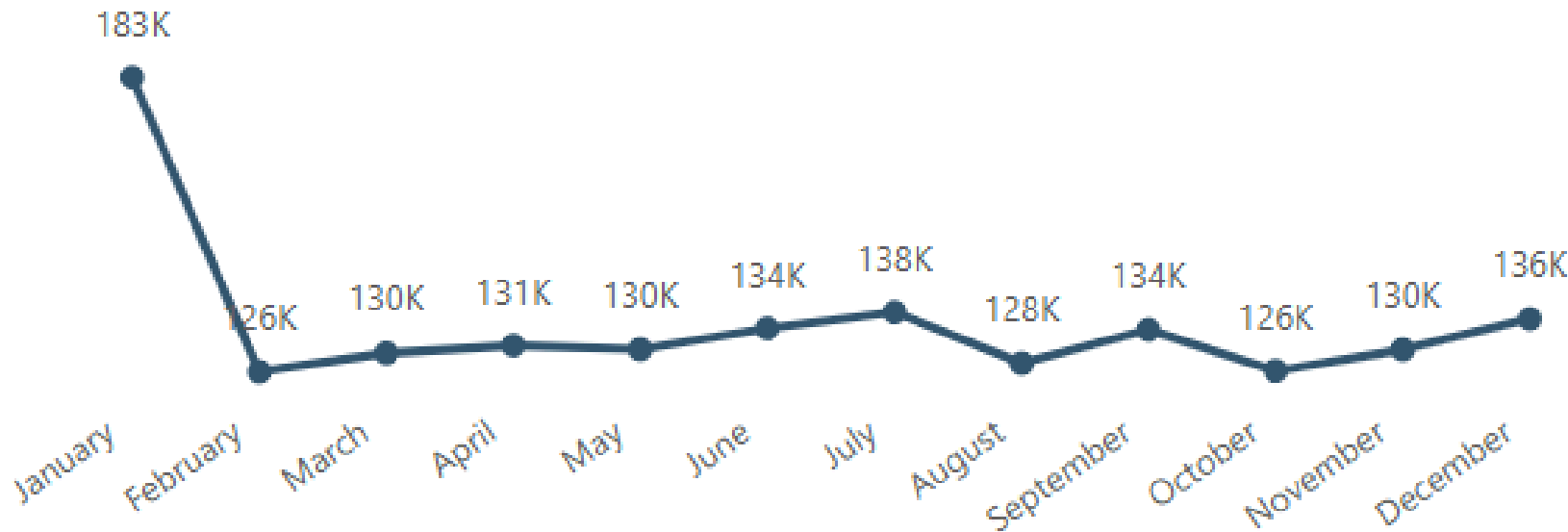
129.40

Average order value

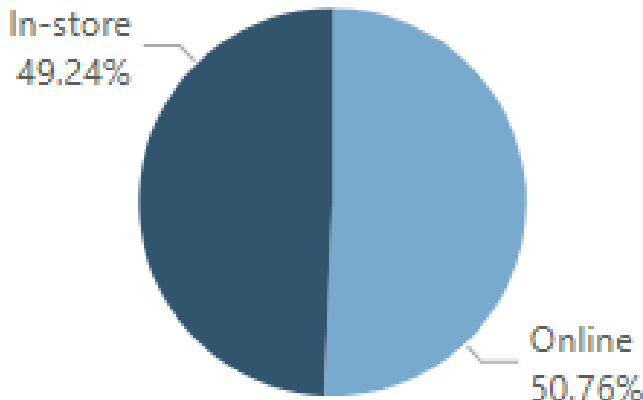
Total Revenue in Different Days



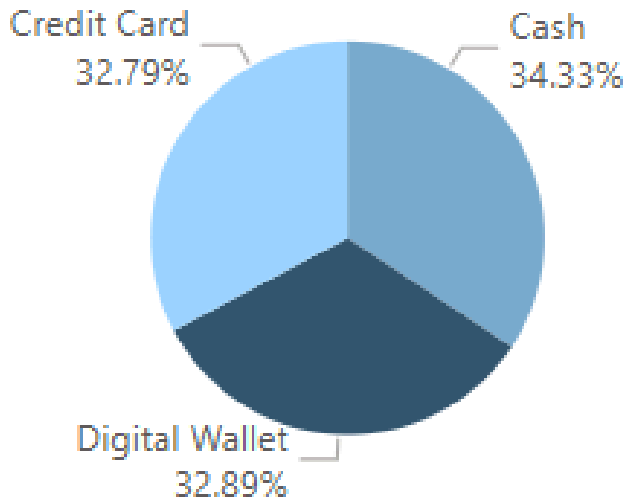
Total Revenue across Different Months



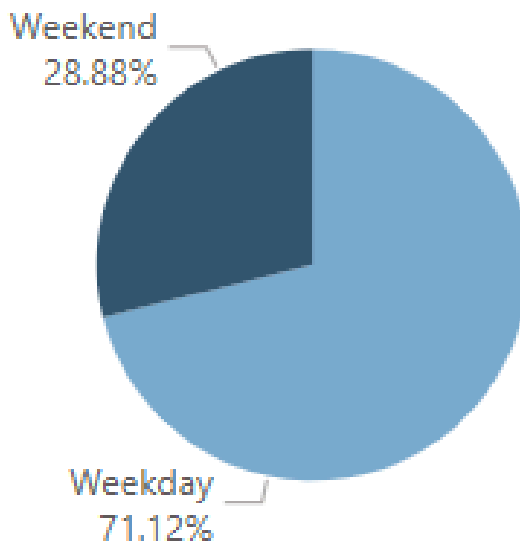
Total Quantity Sold



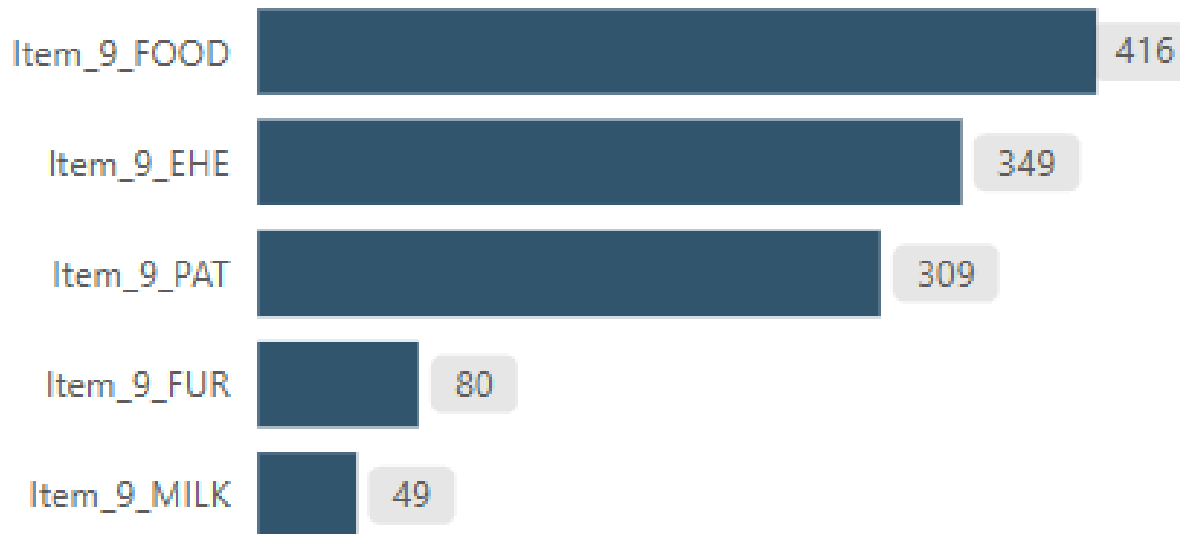
Total Quantity Sold



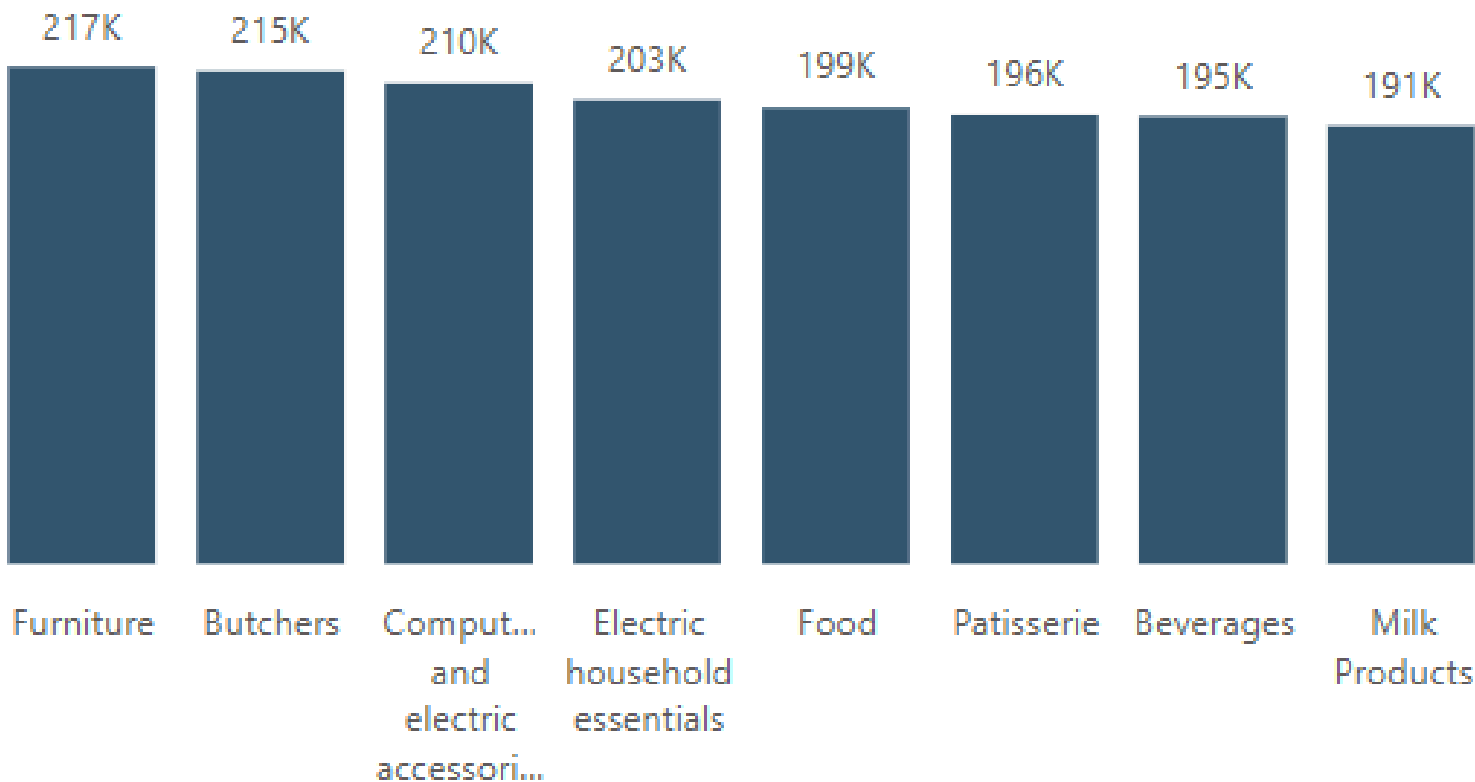
Total Quantity Sold



Top 5 Items Sold



Total Revenue by Category



INSIGHTS

Key Insights:

- Total Transactions: 13K+
- Total Revenue Generated: \$1.63M
- Total Quantity Sold: 69K+ items
- Average Order Value: \$129.40

Business Insights:

- January generated the highest revenue (\$183K)
- 71% of sales happen on weekdays, while 29% occur on weekends
 - Furniture is the top revenue-generating category
 - Most sold items: item9_FOOD and item9_EHE
- Revenue distribution is almost similar across weekdays, showing day of purchase has minimal impact on revenue
- Customers mostly prefer Digital Wallet payments when purchasing Electric Household Essentials



THANK YOU